

PART A [25 marks]

This section consists of 3 questions. Answer **all** questions.

1. Evaluate the following (if exist):

a) $\lim_{x \rightarrow \frac{1}{2}} \frac{1 - 4x^2}{2x - 1}.$ [3 marks]

b) $\lim_{x \rightarrow 1} \frac{\sqrt{x} - 1}{x^2 - 1}.$ [3 marks]

c) Evaluate $\lim_{x \rightarrow -\infty} \frac{\sqrt{2x^2 + 3x}}{5x + 1}.$ [3 marks]

2. Consider the parametric equation of the curve

$$x = \cos^3 \theta \quad \text{and} \quad y = \sin^3 \theta, \quad 0 < \theta \leq 2\pi.$$

a) Find $\frac{dy}{dx}$ and express your answer in terms of θ . [4 marks]

b) Find the value of $\frac{dy}{dx}$ if $x = \frac{\sqrt{2}}{4}$. [3 marks]

c) Show that $\frac{d^2y}{dx^2} = \frac{1}{3\cos^4 \theta \sin \theta}$. Hence, calculate $\frac{d^2y}{dx^2}$ at $\theta = \frac{\pi}{3}$.

[4 marks]

3. Find the relative extremum of the curve $y = x^3 - 4x^2 + 4x$. [5 marks]

PART B [75 marks]

This section consists of 7 questions. Answer **all** questions.

- 1 a) If $z_1 = 4 - i$ and $z_2 = 1 - 2i$, find $z_1 - \frac{5}{z_2}$. Express the answer in polar form.

[6 marks]

- b) Simplify $\sqrt{50} + \sqrt{8} + \sqrt{32}$

[4 marks]

- 2 a) Given that $\log_2 x + 2 \log_4 y = 3$ show that $xy = 3$. Hence, solve for x and y which satisfy the simultaneous equation

$$\log_{10}(2x + y) = 1 \quad \text{and} \quad \log_2 x + 2 \log_4 y = 3.$$

[7 marks]

- b) Solve the inequality $-\frac{1}{3}|6 - 5x| + 2 \geq 1$.

[4 marks]

- 3 Given $g(x) = \frac{kx + 8}{4x - 5}$, $x \neq \frac{5}{4}$ where k is a constant.

- a) Find the value of k if $(g \circ g)(x) = x$.

[6 marks]

- b) Find the value of k so that $g(x)$ is not a one-to-one function.

[5 marks]

4. The functions f and g are defined as $f(x) = \frac{3x + 4}{x - 2}$, $x \neq 2$ and $g(x) = 3 - x$.

- a) Find $f^{-1}(x)$ and $g^{-1}(x)$.

[4 marks]

- b) Evaluate $(f \circ g^{-1})(3)$.

[3 marks]

- 5 a) Given that

$$f(x) = \begin{cases} \frac{|x^2 - x - 2|}{x^2 - 2x} & , x \neq 0, 2 \\ 0 & , x = 2 \end{cases}$$

Find the $\lim_{x \rightarrow 2} f(x)$. Is $f(x)$ continuous at $x = 2$? [7 marks]

- b) Evaluate $\lim_{z \rightarrow 4} \frac{\sqrt{z}-2}{z-4}$, if it exist. [4 marks]

- 6 a) Differentiate $f(x) = \sqrt{x} + \frac{1}{\sqrt{x}}$ with respect to x . [3 marks]

- b) A curve with equation $x^2 - 3y^2 = ae^{y-2x} + by - 6$, where a and b are constants, passes through the point $(1, 2)$.

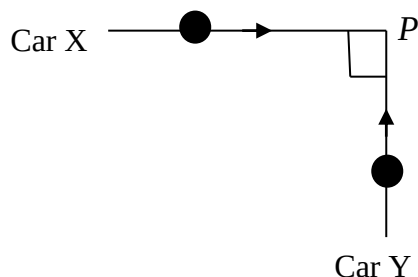
- i) Given $\frac{dy}{dx} = 1$ at $(1, 2)$, determine the values a and b .

[8 marks]

- ii) Evaluate $\frac{d^2y}{dx^2}$ at $(1, 2)$.

[5 marks]

7. Car X is travelling east at a speed of 80 km/h and car Y is travelling north at 100 km/h as shown in the diagram below. Obtain an equation that describes the rate of change of the distance between the two cars. Hence, evaluate the rate of change of the distance between the two cars when car X is 0.15 km and car Y is 0.08 km from P.



[8 marks]

END OF QUESTION PAPER

SUGGESTED ANSWER**PART A**

- 1) a) -2 b) $\frac{1}{4}$ c) $-\frac{\sqrt{2}}{5}$
- 2) a) $\frac{dy}{dx} = -\tan \theta$ b) $\frac{dy}{dx} = -1$ or $\frac{dy}{dx} = 1$ c) $\frac{d^2y}{dx^2} = 6.158$
- 3) $\max : \left(\frac{2}{3}, \frac{32}{27} \right)$, $\min : (2, 0)$

PART B

- 1) a) $3 - 3i$; $3\sqrt{2}(\cos(-0.785) + i \sin(-0.785))$ b) $11\sqrt{2}$
- 2) a) $x = 1, y = 8$; $x = 4, y = 2$ b) $\left\{ x : \frac{3}{5} \leq x \leq \frac{9}{5} \right\}$
- 3) a) $k = 5$ b) $k = -\frac{32}{5}$
- 4) a) $f^{-1}(x) = \frac{2x+4}{x-3}$; $g^{-1}(x) = 3-x$
- b) -2
- 5) a) $\lim_{x \rightarrow 2} f(x)$ does not exist; discontinuous b) $\frac{1}{4}$
- 6) a) $\frac{1}{2\sqrt{x}} - \frac{1}{2\sqrt{x^3}}$ b) i) $a = 5$, $b = -5$ ii) $\frac{d^2y}{dx^2} = -\frac{3}{4}$
- 7) $\frac{dz}{dt} = \frac{x \frac{dx}{dt} + y \frac{dy}{dt}}{z}$; $\frac{dz}{dt} = -117.65 \text{ km/h}$